

Edge Geography is XNLP-hard for Pathwidth and in XP for Tree-Partition Width

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Edge Geography

A classical PSPACE-complete problem.

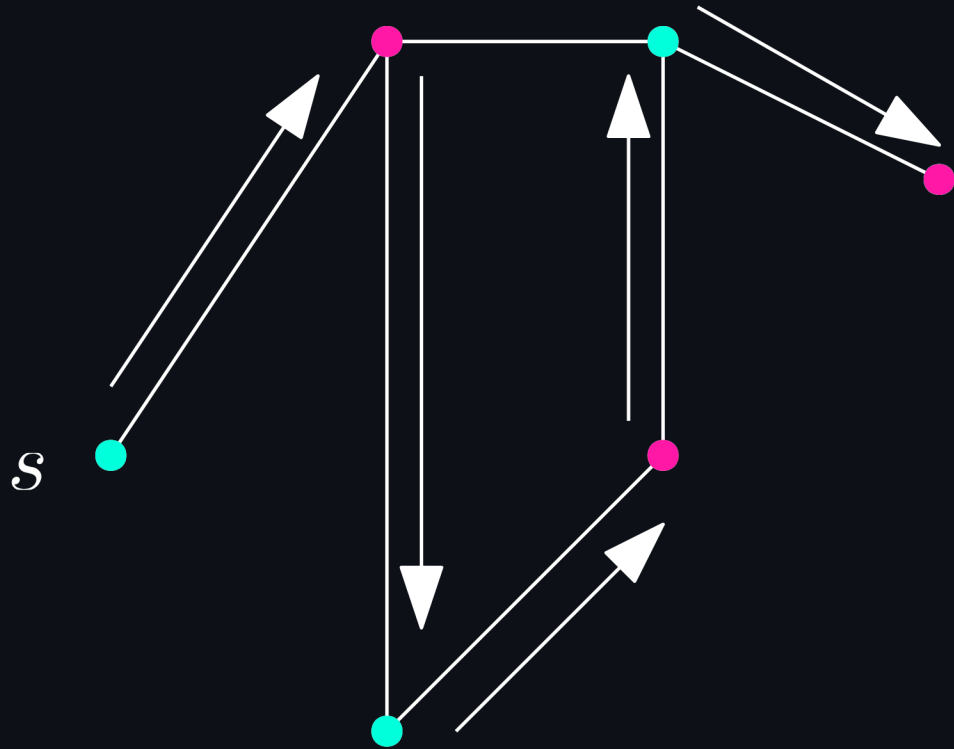
A token starts at a vertex s .

Players alternate:

1. choose an unused edge from the current vertex,
2. move the token along it,
3. delete that edge.

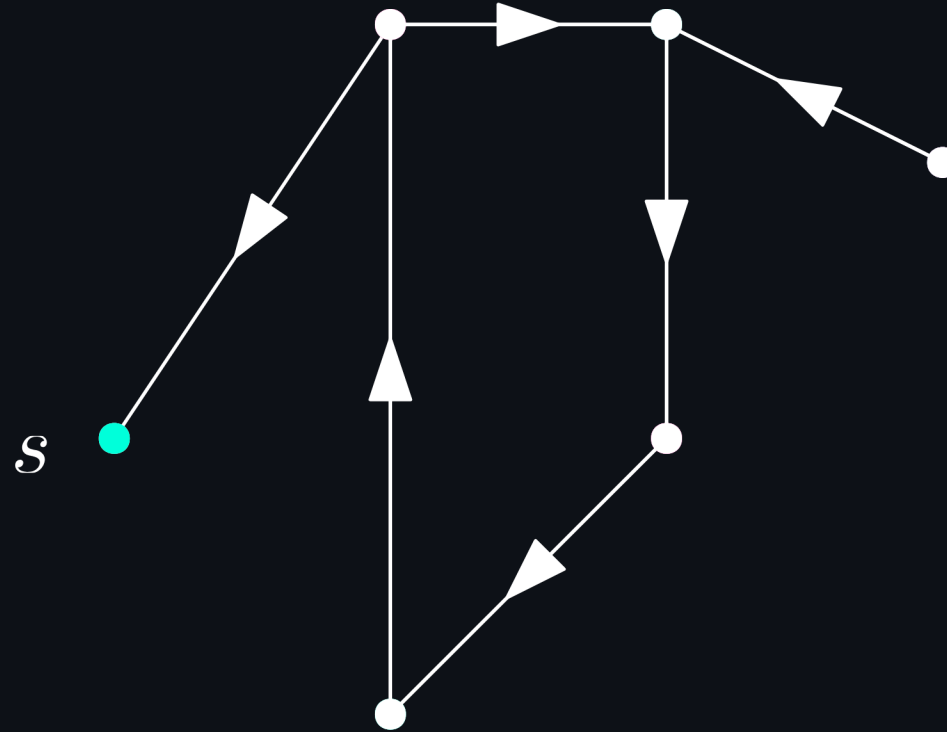
The first player unable to move loses.

Undirected Edge Geography



Player 1 wins!

Directed Edge Geography



Player 2 wins!

Parameterizing the problem

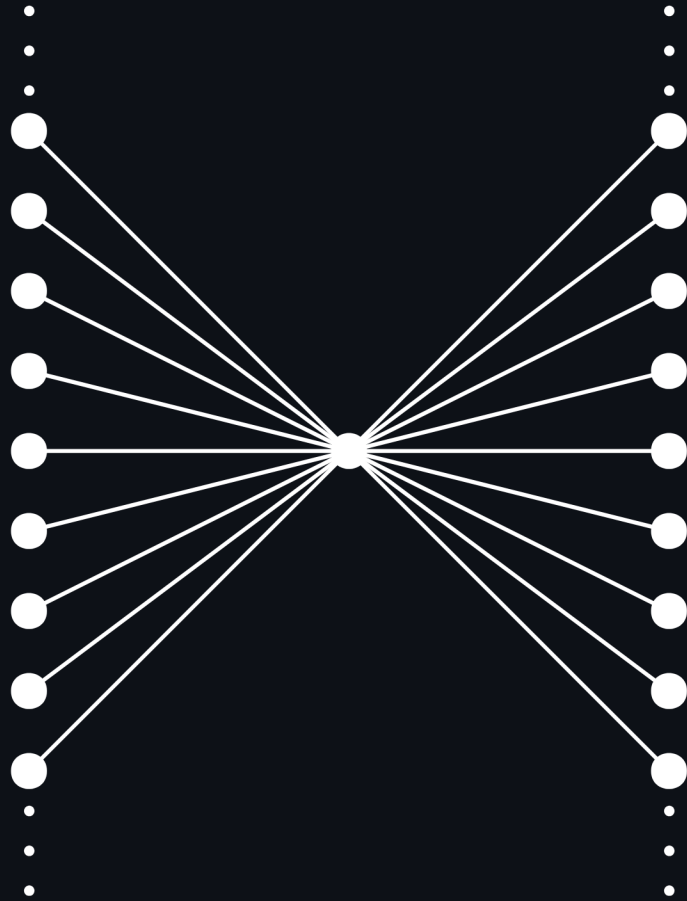
Theorem (Bodlaender, modern language)

(Undirected) Edge Geography is solvable in time $f(k, d) \cdot n$ where k is the treewidth of the graph and d is the maximum degree of any vertex in the graph.

Exponential time becomes "linear time" when two parameters are controlled!

What if we allow d to be unbounded?

Arbitrarily many crossings



when $\Delta(G)$ is unbounded!

Our first result

Theorem

Directed Edge Geography is XNLP-hard parameterized by pathwidth.

Existence of FPT algorithm would imply

$$\text{FPT} = \text{W}[1] = \dots = \text{W}[t] = \dots = \text{W}[\text{SAT}] = \text{W}[\text{P}] = \text{XNLP}.$$

Bounded pathwidth $\Rightarrow$ bounded treewidth.

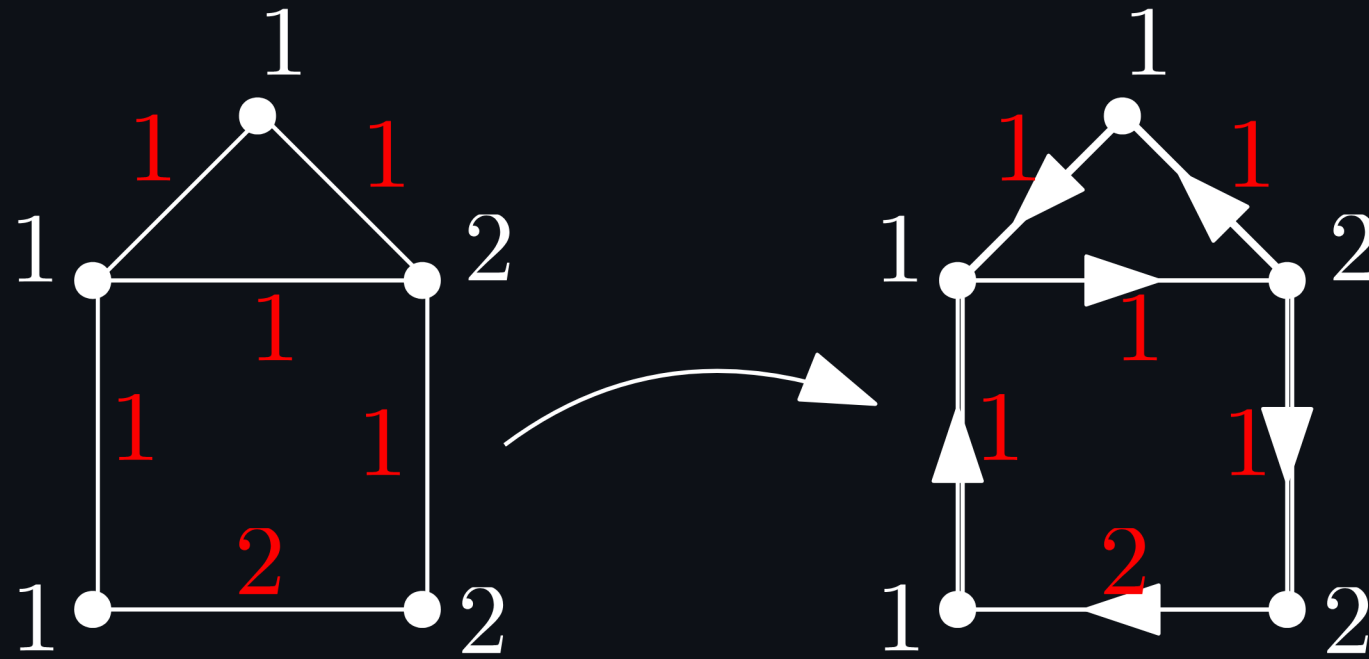
Pathwidth, and thus treewidth, is (given all reasonable assumptions) not enough!

This result transfers to Undirected Edge Geography in a simple way!

Source problem: Chosen Maximum Outdegree

Does there exist an orientation of every edge so that each vertex respects its budget.

$$\sum_{e \in \theta_w(v)} w(e) \leq t(v) \quad \forall v$$



Known to be XNLP-complete when parameterized by pathwidth.

Reduction idea in two phases

Phase I — Player 1 encodes an orientation

For each input edge $e = \{x, y\}$, Player 1 chooses one of two sides.

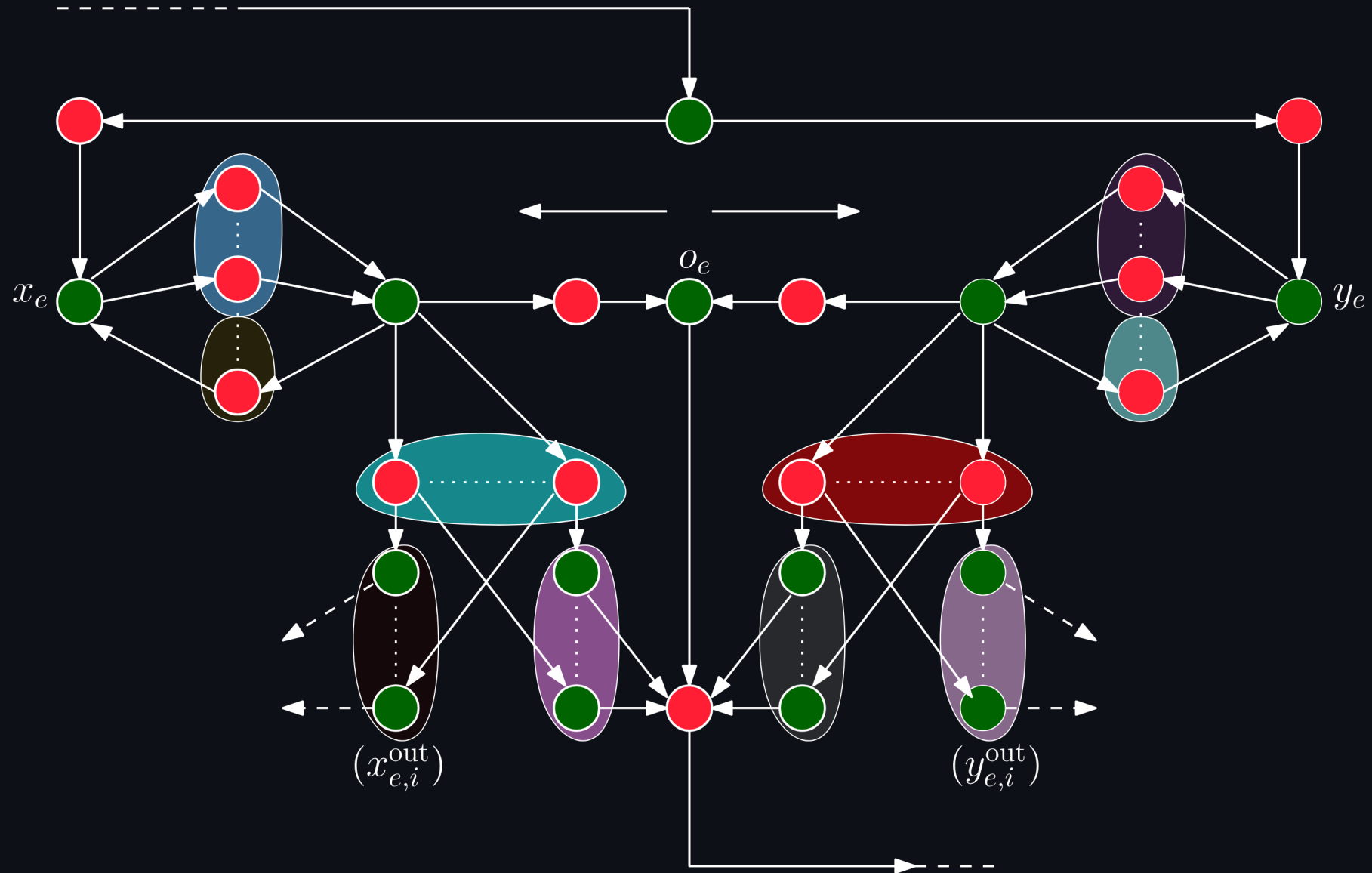
$$\text{choose the } y\text{-side} \iff x \rightarrow y$$

Phase II — Player 2 challenges a vertex

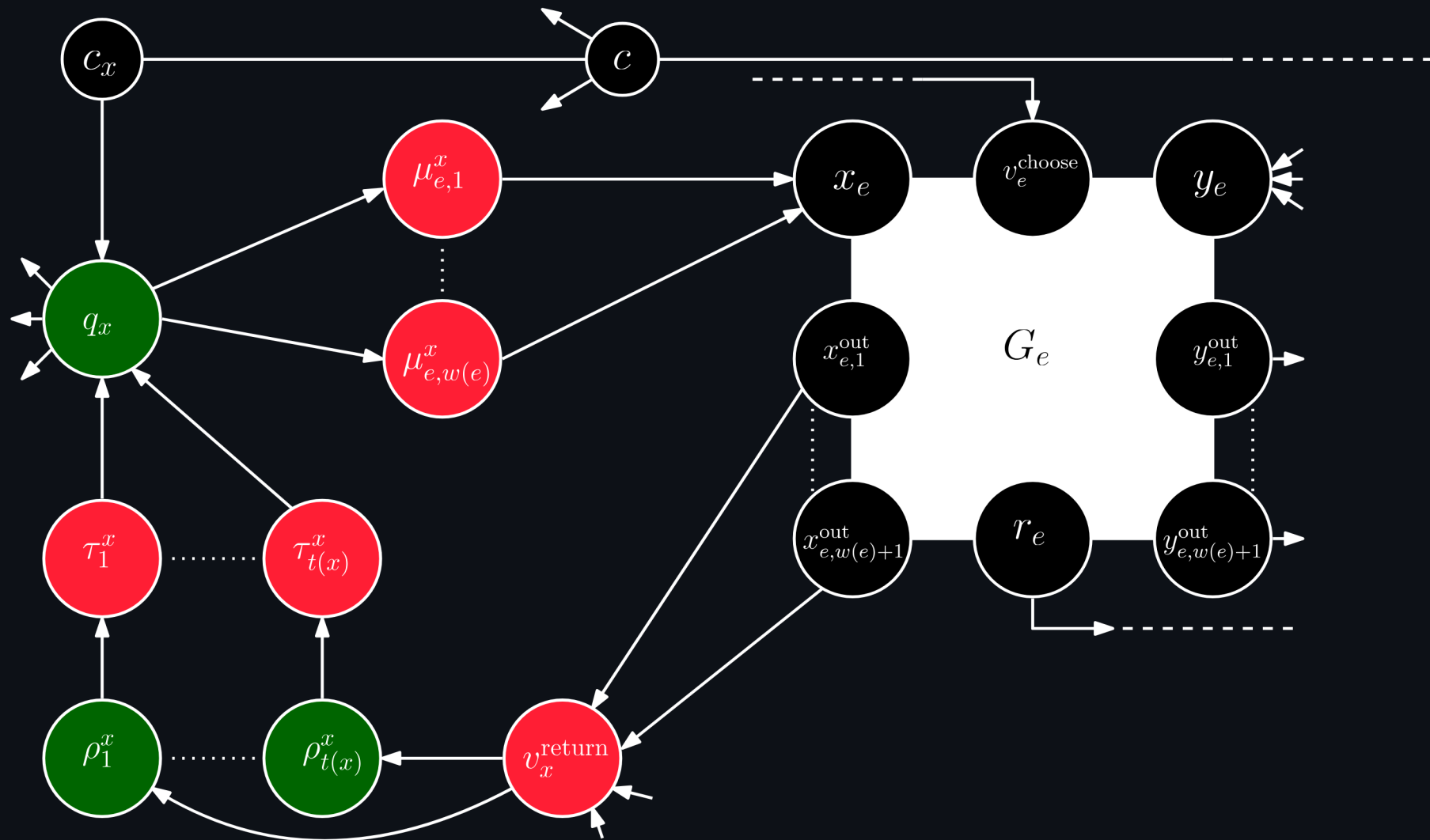
Player 2 chooses x and tests the total weight oriented out of x .

If outgoing weight $> t(x)$, Player 2 has one excursion too many.

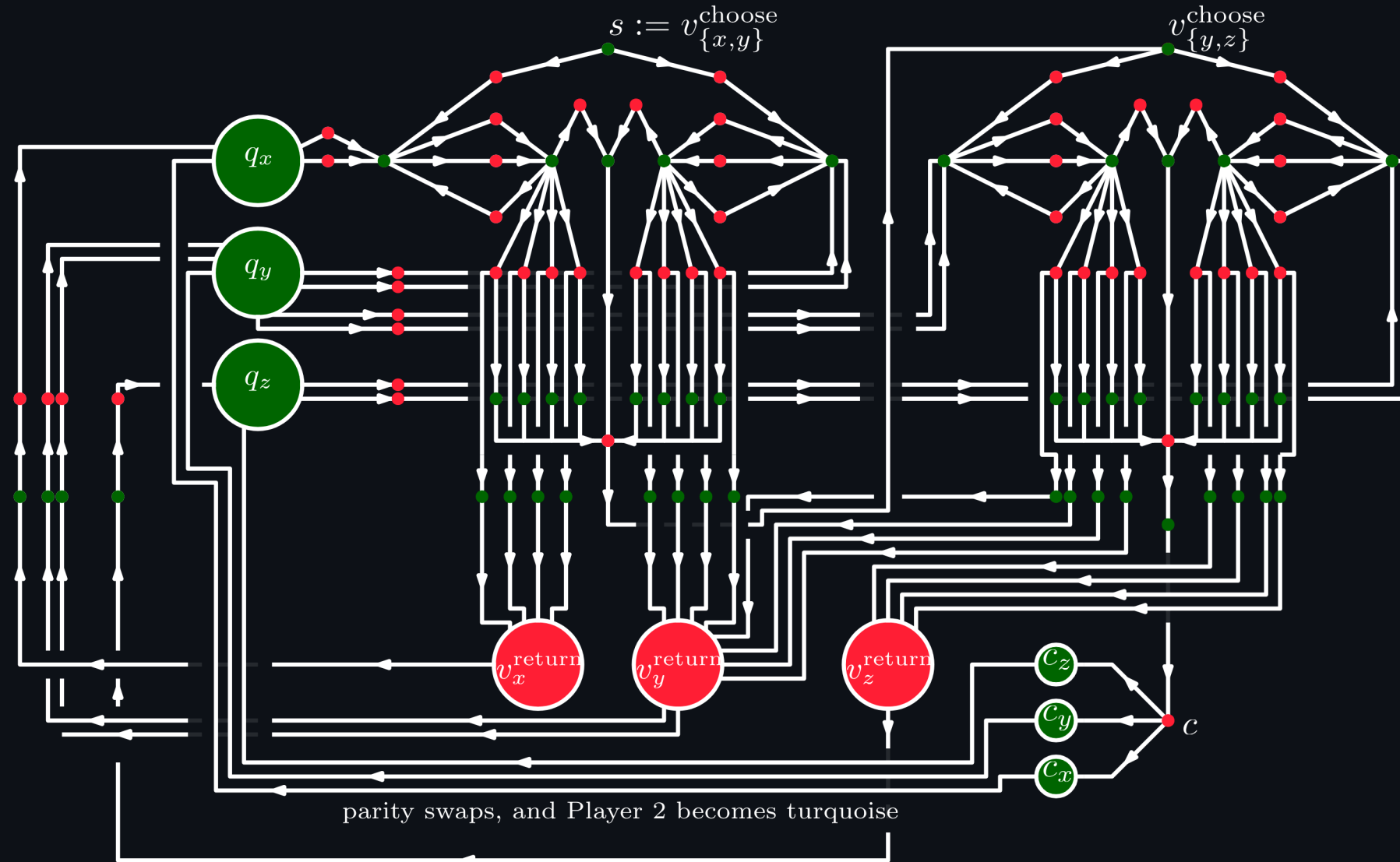
Phase 1



Phase 2



The full reduction for a small source instance



Why correctness follows

Feasible orientation

$$\sum_{e \in \theta_\omega(x)} w(e) \leq t(x) \text{ for every } x.$$

Player 2 eventually runs out of non-losing challenges.

Player 1 wins

Infeasible orientation

For some x ,

$$\sum_{e \in \theta_\omega(x)} w(e) > t(x).$$

Player 2 chooses this x and survives one challenge too many.

Player 2 wins

Why pathwidth is preserved

Process an input edge $e = \{x, y\}$ when x and y coexist in the source path decomposition.

Keep only the necessary anchors alive.

Introduce auxiliary gadget vertices locally, then forget them immediately.

$$\text{pw}(G') = O(k)$$

Directed to undirected

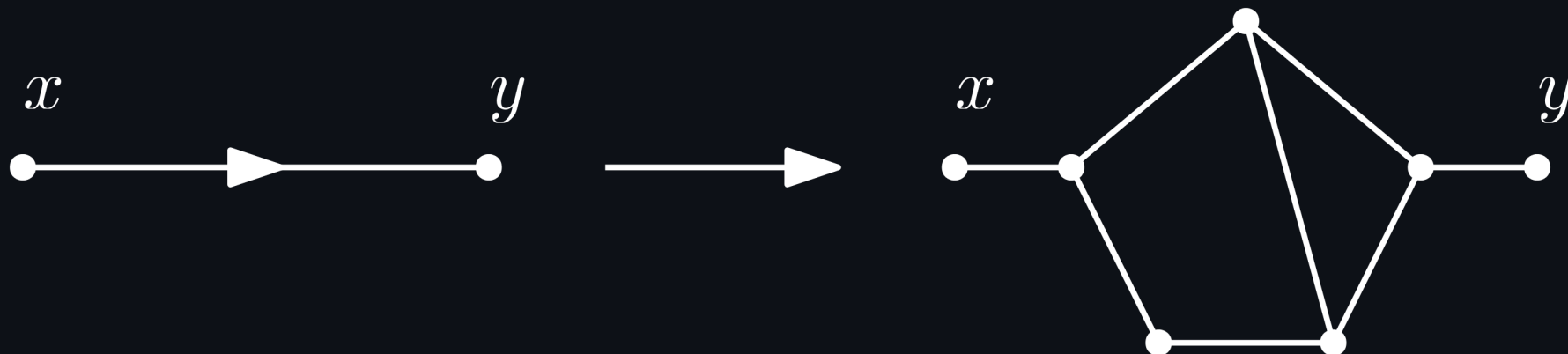
Each directed edge is replaced by a constant-size pseudoarc.

$$u \rightarrow v \quad \rightsquigarrow \quad \text{undirected pseudoarc}$$

Safe traversal works from u to v .

Entering the wrong way is losing under optimal play.

Directed hardness transfers to Undirected Edge Geography.



XP membership for width of treepartition

A rooted tree partition groups vertices into bags.

Edges are either:

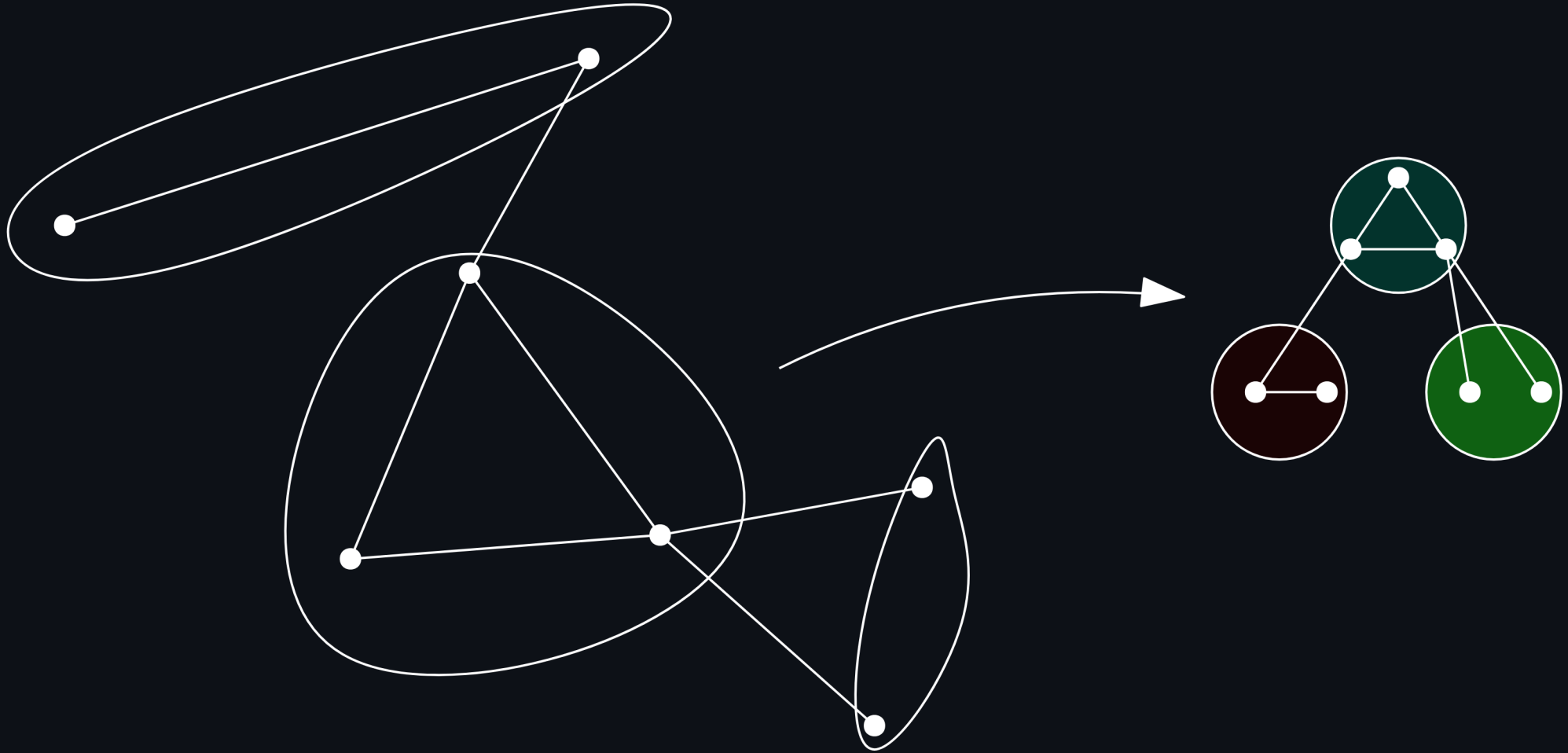
- inside one bag, or
- between adjacent bags of the tree.

For width k , a parent-child cut has at most k^2 ports (on simple graphs).

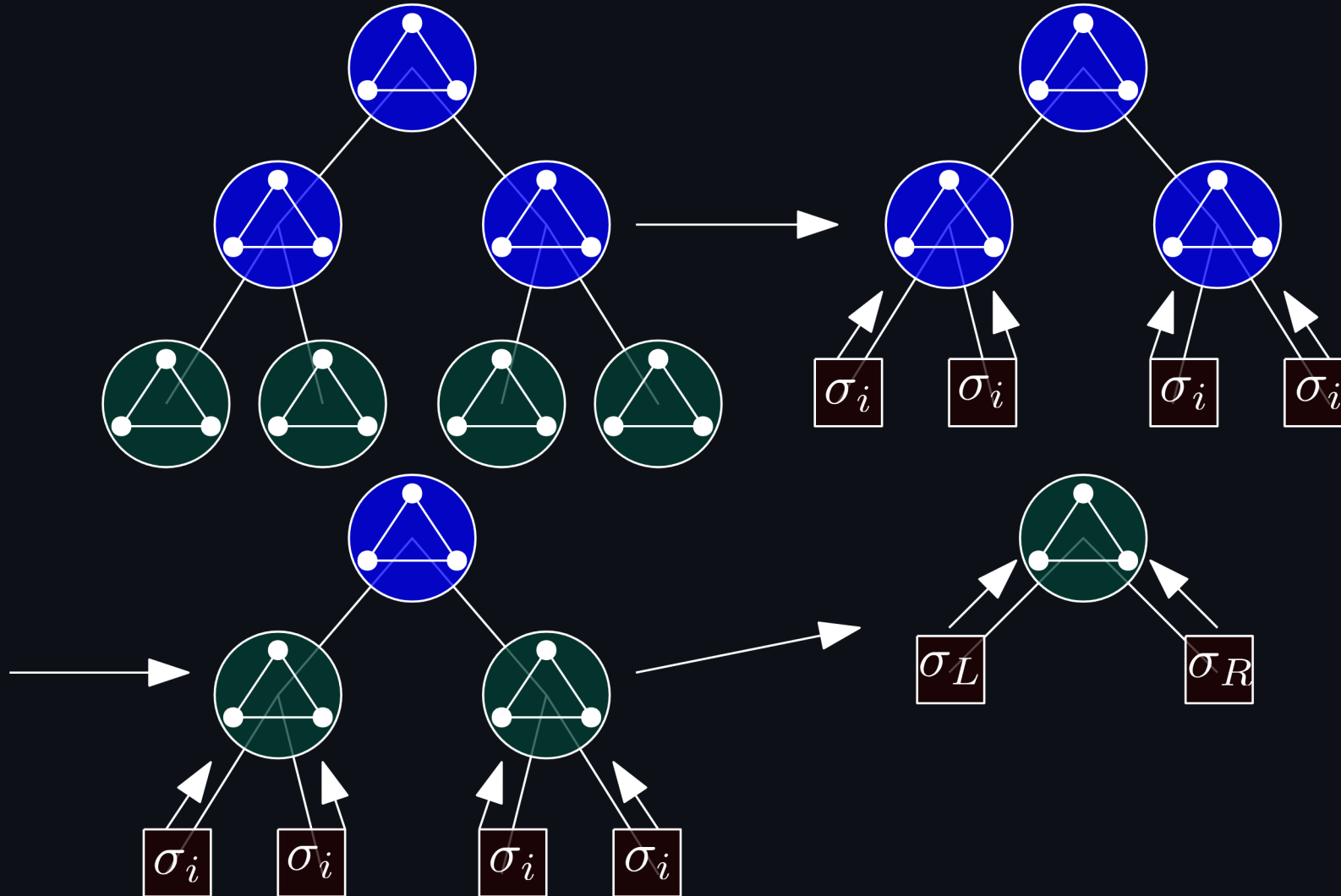
Theorem

Undirected Edge Geography $\in \text{XP}$ when parameterized by tree-partition width.

Tree-partition

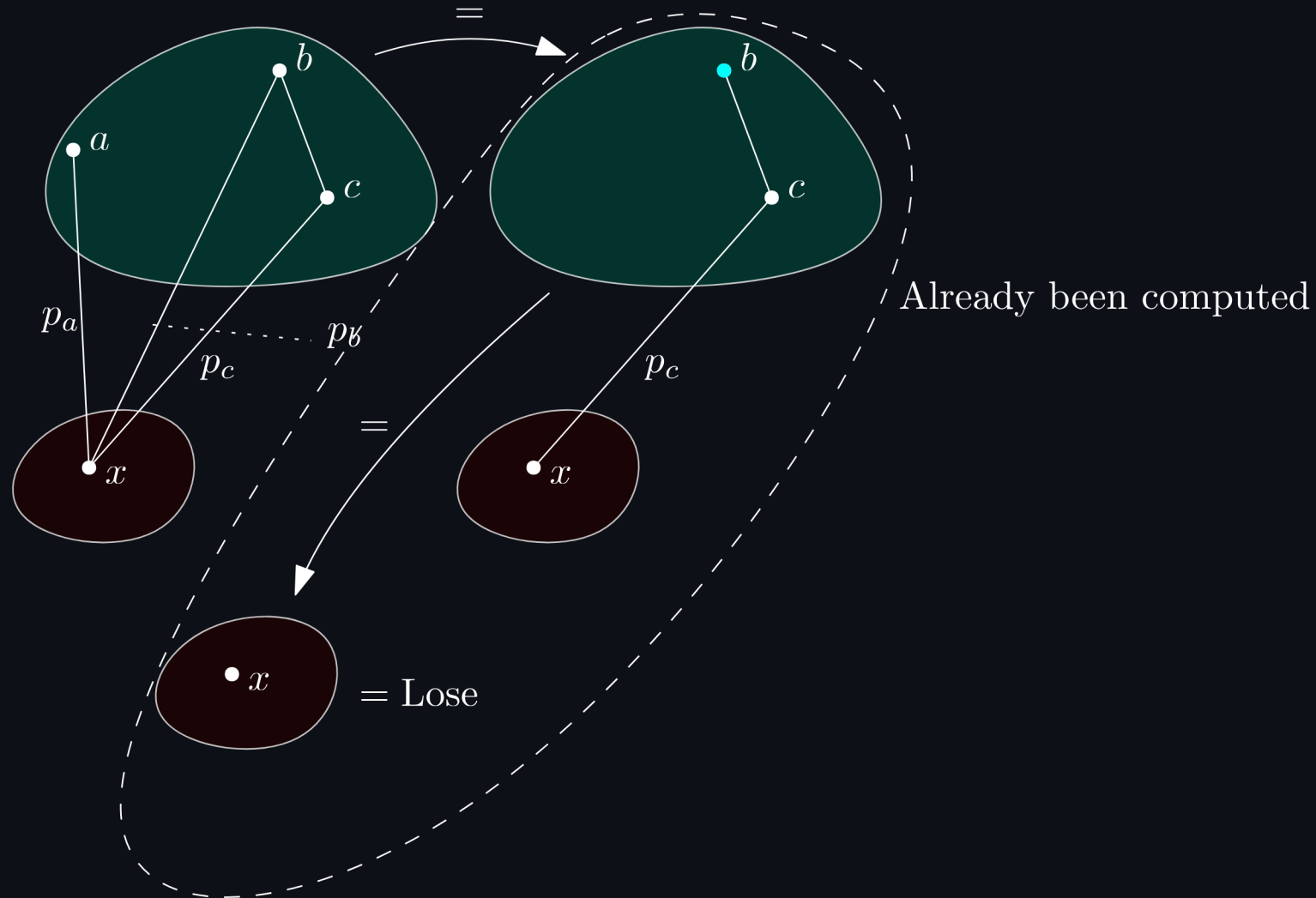


A rough overview of the algorithm



A solved child conveys a boolean function

We want to know: If I move from a into p_a , what happens?



Child counters comprise the worst factor

The n -dependent part

For fixed k , there are t_k possible child types.

Each counter lies in $\{0, \dots, n\}$, so

$$m \in \{0, \dots, n\}^{t_k} \implies \#m \leq (n+1)^{t_k} = n^{O(t_k)}.$$

The other factors

Unused ports and internal edges contribute

$$2^{k^2} \cdot 2^{\binom{k}{2}} = 2^{O(k^2)}.$$

All other boundary data depends only on k ; collect it into $g(k)$.

$$\underbrace{n}_{\text{bags}} \cdot \underbrace{g(k)}_{\text{includes } 2^{O(k^2)}} \cdot \underbrace{n^{O(t_k)}}_{\text{counter states}} = n^{f(k)}.$$

Ridiculously bad in practice

Let T_r be the number of ambient types on an interface with r ports.

ports r	0	1	2	3	4
ambient types T_r	1	2	676	$\approx 2^{768}$	$\geq 2^{2^{4058}}$

Illustrative ambient count obtained by unwinding Definition 35.

The number of realizable types may be vastly smaller.

A width- k cut may have k^2 ports.

The crude bound is a tower of exponentials of height $O(k^2)$.

Takeaway

Pathwidth

small separators

uncontrolled residual history

XNLP-hard

Tree-partition width

bounded interfaces

residual play can be compressed

XP

Is Edge Geography in XP parameterized by pathwidth alone?
Is Edge Geography XALP-hard when parameterized by treepartition width?